

EXCAVATION AND TRENCH SAFETY

Control collapse, underground utilities and heavy equipment hazards.

PURPOSE AND SCOPE

Prevent trench collapse, utility strikes, falls and equipment hazards during excavation work. Verify permits and underground utilities before excavation.

KEY HAZARDS

 CAVE-IN	 UTILITY STRIKE	 FALL	 WATER
 ATMOSPHERE	 EQUIPMENT	 FALLING MATERIAL	 ACCESS

CRITICAL VALUES TO VERIFY IN THE FIELD

1.52 m

General protective-system trigger

7.6 m

Maximum lateral travel to ladder

0.61 m

Minimum spoil setback



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

EXCAVATION AND TRENCH SAFETY

Control collapse, underground utilities and heavy equipment hazards.

STANDARDS AND FIELD REQUIREMENTS

OSHA 1926 Subpart P requires competent-person inspection, utility location, access, protective systems, water and atmosphere controls.

OSHA 1926.651(b)

UNDERGROUND UTILITIES

Locate and mark all services before excavation and verify them by a safe method when approaching.

OSHA 1926.651(c)

SAFE ACCESS

In trenches 1.22 m or deeper, keep workers within 7.6 m lateral travel of a ladder.

OSHA 1926.652(a)

PROTECTIVE SYSTEM

At 1.52 m or deeper, provide sloping, shoring or shielding unless the excavation is entirely stable rock.

OSHA 1926.651(k)

COMPETENT-PERSON INSPECTION

Inspect daily before work and again after rain or any condition change.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Excavation permit and layout | ☐ Utility-detection record |
| ☐ Soil classification | ☐ Shoring / sloping / shield design |
| ☐ Daily competent-person inspection | ☐ Emergency and dewatering plan |

SUPERVISOR DISCUSSION POINTS

- Is the permit approved?
- Are utilities identified?
- Is trench protection installed?

TRENCH PROTECTIVE-SYSTEM INSPECTION



Shield / shoring - Ladder - Spoil setback - Barrier - Utility marking



EXCAVATION AND TRENCH SAFETY

Control collapse, underground utilities and heavy equipment hazards.

CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. AVOID OR REDUCE EXCAVATION

Verify permit. Locate underground services.

2. VERIFY SERVICES AND SOIL

Use trench protection. Provide ladders. Barricade excavation.

3. INSTALL PROTECTIVE SYSTEM

Keep spoil away from edge. Maintain equipment clearance. Inspect after rain.

4. CONTROL ACCESS AND CONDITIONS

Gas test if required. Prepare emergency plan.

CRITICAL WARNING



Never enter an unprotected excavation.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|---|
| ☐ Permit approved | ☐ Utilities identified |
| ☐ Protection installed | ☐ Ladder available |
| ☐ Barricaded | ☐ Ground stable |
| ☐ Gas checked | ☐ Equipment clear |
| ☐ Standby ready | |



EXCAVATION AND TRENCH SAFETY

Control collapse, underground utilities and heavy equipment hazards.

ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			